

An investigation of data analysis method for hydraulic fracturing based on the water hammer effect

Jianguo Shen^{a,*}, Huaiyin He^a, Yanchao Li^a, Longqing Zou^a, Yixuan Wang^a, Zhaoying Zhu^b,
Lingkong Guo^b, Shuoran Fu^b

^a Shale Gas Exploration and Development Project Management Department of PetroChina Exploration and Drilling Engineering Company Limited, Chuanqing Branch, Chengdu, Sichuan, 610052, China

^b College of Energy Resources, Chengdu University of Technology, Chengdu, Sichuan, 610059, China

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ABSTRACT

Hydraulic fracturing plays a crucial role in the development of unconventional resources, and assessing fracture geometry is a fundamental aspect of hydraulic fracturing analysis. Among the various technologies for evaluating fracture geometry, the water percussion signal-based method has gained popularity due to its cost-effectiveness, convenience, and real-time capabilities. In this study, we present a methodology for analyzing fracture data that utilizes the water percussion effect. By leveraging the impact of fractures on water percussion pressure attenuation, we propose an inverse calculation approach to determine fracture geometry. Firstly, we introduce the RCI (Reservoir-Completion-Interaction) circuit model, which effectively addresses the bottom hole fracture boundary and simulates the variation of water percussion pressure in the wellbore. Secondly, the simulation results are utilized in an iterative process to determine the RCI values, which are then utilized for the inversion calculation of fracture geometry. Finally, we apply this method to a field case and compare the simulation results with microseismic monitoring data. A larger resistance (R) value indicates a smaller fracture volume, while the (C) value can be used to monitor any poorly performing stages of the fracturing simulation process. The (I) value primarily affects the parameter calculation of the fracture size, especially the fracture width. The results of the field application show that the accuracy rate exceeding 80%, validating the reliability of the model and providing a valuable reference for field fracturing data analysis.

1. Introduction

Unconventional oil and gas development is currently a global focus in the energy industry. Hydraulic fracturing stands as the dominant technology employed for developing unconventional oil and gas reservoirs. However, the geometric shape of fractures resulting from hydraulic fracturing profoundly impact on the ultimate production of these reservoirs [1]. Consequently, the evaluation of hydraulic fracture geometry holds significant importance in hydraulic fracturing diagnosis. Several methods for fracture monitoring have been developed, including microseismic monitoring technology [2] and distributed fiber optic monitoring technology [3]. These methods can depict the distribution characteristics of fractures [4] and assess the effectiveness of fracturing. However, they are constrained by operational challenges, complex construction requirements, and substantial investments [5]. Therefore, there is a need for cost-effective methods to evaluate fracture geometry.

The fracture geometry diagnosis technique based on water percussion signal is a promising method, with the advantages of high efficiency, easy implementation and high real-time performance. Many studies have demonstrated the feasibility of using pressure waves to diagnose fracture geometry.

In this research, a water hammer model based on circuit principles is established to calculate fracture geometry after hydraulic fracturing. The water percussion model is employed to simulate and analyze the changes in water percussion pressure in the wellbore. By employing an iterative methodology, we ascertain optimal fracture characteristic parameters and compute their corresponding geometric counterparts. To validate the model's accuracy, the fracture geometry obtained from the model is compared to the geometry interpreted through microseismic analysis. We hope this research can provides theoretical guidance for optimizing hydraulic fracturing to enhance shale gas recovery.

* Corresponding author.

E-mail address: 328599896@qq.com (J. Shen).

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2. Methodology

The traditional water hammer mathematical model [6] adopts a constant pressure boundary condition, which does not match the actual engineering situation. By analyzing the pressure curve fluctuations on-site, it becomes evident that the pressure rapidly diminishes to zero following the cessation of pumping. This occurrence can be attributed to energy dissipation resulting from hydraulic fractures. A new boundary condition based on circuit principles is established [7], where the bottom-hole fracture is regarded as a closed circuit. This assumption effectively simulates the pressure variations experienced by bottom-hole fractures, which offers the following advantages.

- (1) flow resistance, fluid compressibility, and inertial effects are considered;
- (2) boundary condition is dynamic;
- (3) it can be easily coupled with the wellbore water hammer equation.

2.1. Characteristics calculation

The fundamental equations required for establishing the water percussion pressure model are the continuity equation for water percussion and the equation of motion:

$$\frac{\partial H}{\partial t} + v \sin \theta + \frac{a^2}{g} \frac{\partial v}{\partial x} = 0 \quad (1)$$

$$g \frac{\partial H}{\partial x} + \frac{\partial v}{\partial t} + \frac{f}{2D} |v| v = 0 \quad (2)$$

Pressure wave velocity in wellbore [8] is expressed as:

$$a = \sqrt{\frac{E/\rho}{1 + (1 - \nu^2) \frac{D}{e} \frac{E}{E_0}}} \quad (3)$$

where, H is the head of the pressure measuring tube, m; f is the along-range resistance coefficient; θ is the inclination of the pipe; v is the liquid velocity, m/s; E is the liquid elasticity coefficient, Pa; E_0 is the pipe elasticity coefficient, Pa; e is the pipe wall thickness, mm; ν is the Poisson's ratio of the pipe.

However, Eqs (1) and (2) cannot be solved directly and require integration operations by different numerical methods, such as the characteristic line method [9,10], the finite difference method [11,12], and the finite volume method [13,14]. The characteristic line method is widely employed for solving water hammer problems due to its convenience, ease of programming [15], and its ability to significantly enhance computational accuracy.

Introducing a pair of eigenvalues to linearly combine the first-order fitted linear hyperbolic partial differential equations to obtain the ordinary differential equations, denoted as C^+ , C^- , respectively.

$$C^+ : \frac{\partial p}{\partial t} + \rho a \frac{\partial v}{\partial t} + \rho g v \sin \theta = 0 \quad (4)$$

$$C^- : \frac{\partial p}{\partial t} + \rho a \frac{\partial v}{\partial t} - \rho g v \sin \theta = 0 \quad (5)$$

However, these ordinary differential equations do not hold everywhere in the x - t plane, but only on lines with slopes of $1/a$ and $-1/a$, which are called characteristic lines, as shown in Fig. 1.

The ordinary differential equation is integrated along the characteristic line and transformed into a finite difference format as follows (a detailed derivation of the steps can be found in the literature [16]):

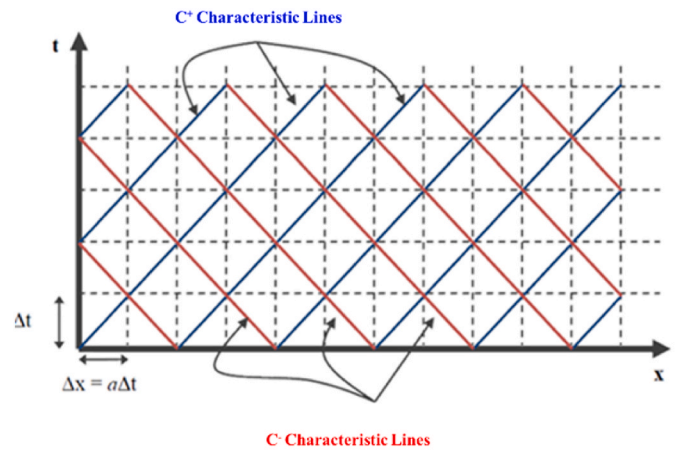


Fig. 1. Characteristic lines grid diagram.

$$\left\{ \begin{aligned} p_i^{j+1} &= \frac{1}{2} (p_{i-1}^j + p_{i+1}^j) + R(v_{i-1}^j - v_{i+1}^j) - S \left(v_{i-\frac{1}{2}}^j - v_{i+\frac{1}{2}}^j \right) \\ v_i^{j+1} &= \frac{1}{2} (v_{i-1}^j + v_{i+1}^j) + T(p_{i-1}^j - p_{i+1}^j) - U \left(v_{i-\frac{1}{2}}^j + v_{i+\frac{1}{2}}^j \right) \\ v_{i-\frac{1}{2}}^j &= \eta v_{i-1}^j + (1 - \eta) v_i^{j+1} \\ v_{i+\frac{1}{2}}^j &= \eta v_i^{j+1} + (1 - \eta) v_{i+1}^j \end{aligned} \right. \quad (6)$$

2.2. Boundary conditions

To construct a mathematical model of water hammer, it is necessary to define boundary conditions at both the wellhead and bottom hole fracture. The wellhead boundary primarily captures the changing conditions of fluid pressure and flow rate at the wellhead following the cessation of fracturing pumping. Typically, a staged pump shutdown is employed after completing the hydraulic fracturing process to prolong the pump stopping time and mitigate potential damage to the wellbore or formation resulting from water hammer effects. The relationship between flow rate varying at the wellhead can be determined by relating the time-varying flow rate to the time-varying valve opening area. When establishing the fracture boundary, the circuit principle is introduced to facilitate model computations and enable the inversion of fracture parameters. This approach aids in accurately representing the hydraulic behavior of the fractures and enables efficient analysis of the water hammer phenomenon.

2.2.1. Well bottom fracture boundary

The fracture is defined as a closed circuit consisting of three elements in series: resistance R , capacitance C and inductance I , as in Fig. 2.

The elements that constitute the fracture boundary at the bottom of the well have different but corresponding physical meanings in the circuit model and the fracturing model, respectively, as shown in Table 1.

R , C and I are all coefficients related to the pressure and flow rate in the fracture, and the resistance R represents the resistance.

$$R = \frac{p + \rho g z + \frac{1}{2} \rho v^2}{v} \quad (7)$$

The capacitance C is defined as the weakness and is the ratio of the fracture volume to the change in pressure potential.

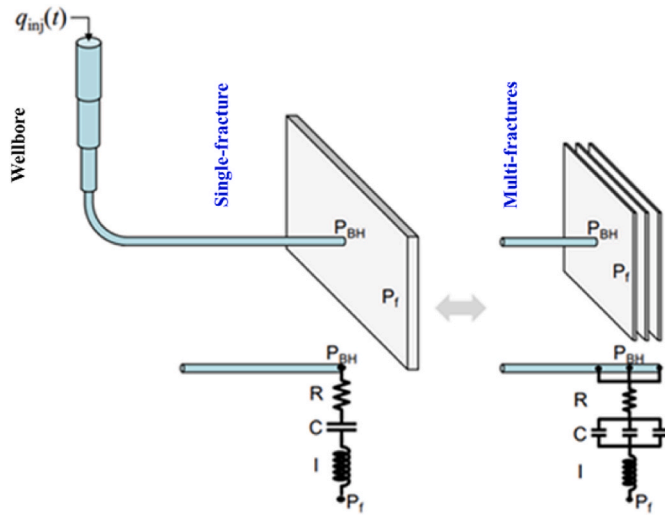


Fig. 2. Schematic diagram of fracture boundary.

Table 1

Table of physical meanings of the elements.

Symbols	Circuits	Fracturing model
R	Resistance	Resistance (near wellbore friction)
C	Capacitance	Fracture weakness
I	Inductors	Inertia
p	Voltage	Pressure potential
v	Current	Flow Rate

$$C = \frac{\Delta V}{\Delta P} \quad (8)$$

The inductance I is inertial and is related to the mass of the fluid.

$$I = \frac{1}{\Delta P} \frac{dQ}{dt} \quad (9)$$

Connecting the elements R (resistor), C (capacitor), and I (current source) in series fulfills the conditions that describe a downhole fracture. This arrangement is suitable because, during water hammer events, the pressure and flow dynamics are primarily governed by the volume of fluid and the characteristics of the fracture itself, assuming a linear flow state within the fracture [17–21]. In a series circuit configuration, current flows into the circuit until the capacitor is fully charged. Subsequently, current will only flow when there is an increase in potential difference or a decrease in resistance, which is analogous to the movement of fluid in a hydraulic fracture.

2.2.2. Fracture size inversion

After determining the values of R, C and I, the seam half-length L_f , seam height h_f and seam width $\bar{w}$ of the fracture can be performed inversely. The difference between the average pressure in the fracture and the horizontal ground stress is called the average net pressure $\overline{\Delta P_0}$.

$$\overline{\Delta P_0} = (P_{BH} - \Delta P_{nwf}) - S_{H \min} \quad (10)$$

$$\Delta P_{nwf} = RQ_0 \quad (11)$$

In the above equation, P_{BH} is the bottomhole pressure, ΔP_{nwf} is the near-well frictional pressure drop, and the average pressure in the fracture is the difference between the bottomhole pressure and the near-well frictional pressure drop. Relating the average net pressure to the fracture volume change [22], the relationship is obtained as

$$\overline{\Delta P_0} = \frac{4}{\pi^2} \frac{E}{1 - \nu^2} E(m) \bar{w} \left(\frac{1/L_f}{2/h_f} \right) \quad (12)$$

Assuming that the change in fracture volume caused by fracture flexibility is only the change in fracture width, the capacitance C can be rewritten as

$$C = \frac{\Delta V}{\Delta P} = \frac{\Delta w h_f L_f}{\Delta P} \quad (13)$$

Substituting Eq (12) into Eq (13), we get

$$C = \frac{\pi^2 h_f L_f}{4E E(m)} \left(\frac{L_f}{h_f/2} \right) \quad (14)$$

Similarly, the equation for inertia I is obtained as

$$I = \frac{\rho L_f}{\bar{w} h_f} = \frac{4\rho E E(m) L_f}{\Delta P_0 h_f} \left(\frac{1/L_f}{2/h_f} \right) \quad (15)$$

where, $E' = \frac{E}{1 - \nu^2}$. Jointly (14) and (15) can derive the fracture size equation.

$$L_f = \sqrt{\frac{CI \Delta P_0}{\rho}} \quad (16)$$

$$h_f = \begin{cases} \frac{4E E(m) C}{\pi^2 L_f^2} & \text{if } (2L_f/h_f) < 1 \\ \sqrt{\frac{8E E(m) C}{\pi^2 L_f}} & \text{if } (2L_f/h_f) \geq 1 \end{cases} \quad (17)$$

$$\bar{w} = \frac{\rho L_f}{I h_f} \quad (18)$$

In practical engineering, segmented fracturing is commonly used for shale gas wells, and each fractured section has more than one cluster of fractures. In this paper, multiple fracture clusters are approximated as a single biplane fracture, and the calculated fracture parameters are equivalent to the dimensional results of the actual fracture network.

As shown in Fig. 3(a) and Fig. 3(b), the boundary condition of a multicluster fracture is considered as a parallel circuit with the following fracture dimensions for the inverse performance

$$L_{fE} = \sqrt{\frac{I_T}{\rho} \sum_{i=1}^n C_i \Delta P_{0i}} = \sqrt{\sum_{i=1}^n L_{fi}^2} \quad (19)$$

$$\bar{w}_E = \sqrt{\sum_{i=1}^n \bar{w}_i^2} \quad (20)$$

3. Field applications

This paper uses MATLAB software for programming the following technical route for shale fracturing stopping pump pressure interpretation based on a mathematical model of water hammer.

- (1) Input parameters. Inputting wellbore parameters, reservoir and fluid parameters, and fluid injection parameters when pumping is stopped.
- (2) RCI iterative fitting.
- (3) Calculation of parameters related to pumping stoppage pressure.
- (4) Fracture parameter inversion.

The program can be used to compare the variation of pressure curves and analyze the influencing factors of water hammer effect. Furthermore, it finds application on construction sites for analyzing and evaluating the impact of hydraulic fracturing in shale horizontal wells.

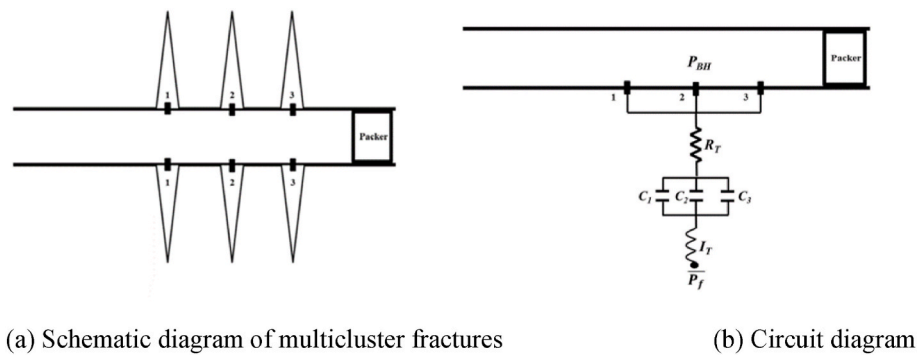


Fig. 3. Schematic diagram of the multicluster fracture circuit (a) Schematic diagram of multicluster fractures (b) Circuit diagram.

3.1. Sensitivity analysis

Utilizing the water hammer pressure model, it becomes possible to analyze the effects of different variables and derive sensitivity factors that influence the variation of water hammer pressure waves. In this study, sensitivity analysis is conducted by varying RCI parameters, as well as wellbore and fluid parameters such as section length, casing size, and fluid viscosity, to assess their impact on water hammer effects.

3.1.1. RCI parameter impact analysis

In accordance with the technical approach of the stop-pump pressure interpretation method, the primary factor for adjusting the simulation results to align with the actual field curve is the modification of RCI values. Since the RCI values are directly associated with the fractures, the impact of RCI parameters on the water-strike response is given priority. To examine this influence, the control variable method is employed, wherein the R, C, and I values are individually varied. By implementing this approach, comparison graphs of water-strike pressure curves are obtained, as illustrated below.

As depicted in Fig. 4, it is evident that a higher resistance value (R) leads to a more rapid decay in the pressure response. This observation suggests a larger frictional pressure drop near the well and a smaller fracture volume. When comparing water hammer pressure curves for different capacitance values (C) (Fig. 5), it can be noted that smaller C values result in a larger discrepancy between the curve and the actual water hammer response. This characteristic can be employed to identify poorly performing stages in the fracturing simulation process. The impact of varying the inertia value (I) on the water hammer response

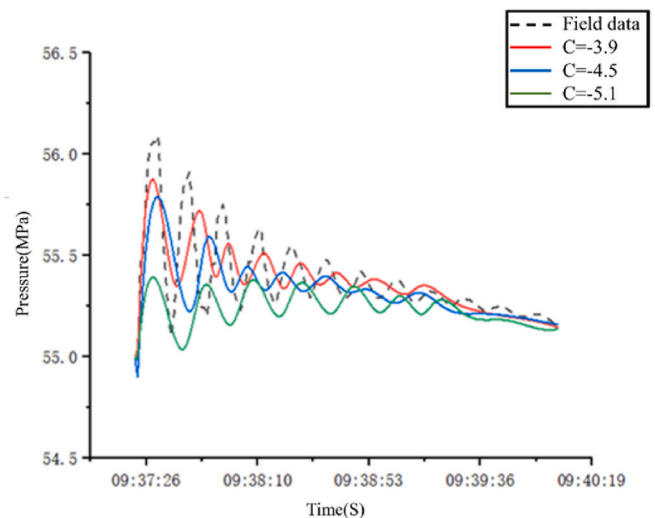


Fig. 5. Pressure curves at different C.

does not exhibit a clear pattern (Fig. 6). However, based on the fracture size equation, it can be inferred that value (I) primarily influences the calculation of fracture size parameters, especially fracture width.

3.1.2. Effect of wellbore and fluid parameters

The construction design of various horizontal wells for achieving

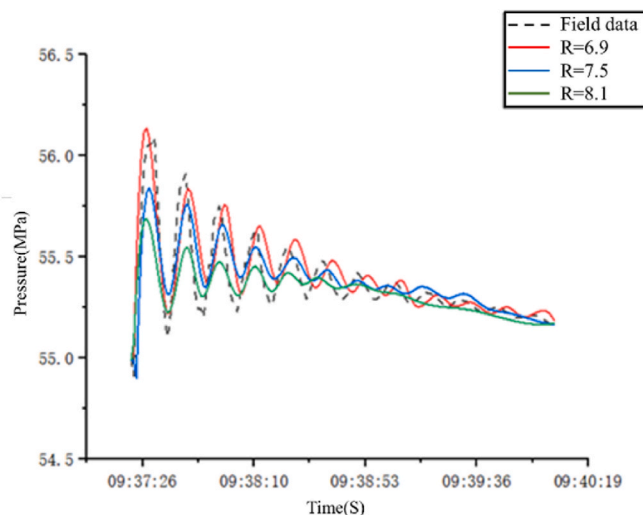


Fig. 4. Pressure curves at different R.

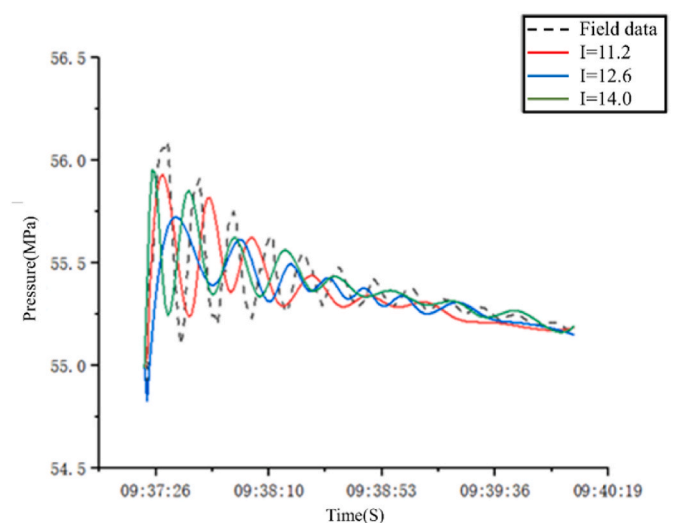


Fig. 6. Pressure curves at different I.

improved fracturing outcomes may vary to some extent based on geological conditions. The key differences primarily lie in variations in section length, casing size, and fracturing fluid viscosity. By comparing the pressure data from different well sections, the impact of wellbore and fluid parameters on the water hammer effect can be analyzed.

The analysis of water hammer pressure variations for different section lengths (Fig. 7) reveals that longer section lengths result in longer pressure waves. This can be attributed to the increased propagation time for momentum within the wellbore. In the case of different casing sizes (Fig. 8), it is observed that the amplitude decreases as the casing size increases. This indicates that an increase in the total fluid volume within the wellbore leads to a reduction in the amplitude of momentum transfer through the wellbore. Furthermore, when varying the viscosity of the fluid (Fig. 9), the analysis demonstrates that higher viscosity leads to faster attenuation of the water hammer signal. This is because higher viscosity fluids experience more rapid dissipation of momentum transfer within the wellbore (see Fig. 10).

3.2. Case analysis

The accuracy of the model was validated through a stopping pressure analysis conducted on shale gas wells located in the W2 well area. The simulation program was utilized to replicate the water hammer effect occurring during pump shutdown, and the resulting fracture parameters were inverted and compared with microseismic monitoring data. This comparison served as a means to verify the accuracy and reliability of the model.

The W2 well area is mainly located in Sichuan Province, where small nose-like fractures are developed and the physical properties are generally good, with abundant shale gas resources. 1 well is a horizontal shale gas well in this area, with the main fracturing section of 2946–4496 m well section, using soluble bridge plugs as the segmentation tool. The construction pressure is 57–67 MPa, the average construction displacement is 14 m³/min, a total of 2353.0t of 100 mesh quartz sand and 734.7t of 40/70 mesh ceramic granules are added, the total fluid used in the fractured layer is 23568.85 m³ (22,828.82 m³ of slick water and 740 m³ of linear rubber), and the average stopping pressure is 45.4 MPa. Figs. 3–7 shows the pressure curve of the fractured layer in this well. The pressure construction curve of one section can be seen more obviously the fluctuation of pressure when stopping the pump. After the construction parameters were input, the iterative R, C and I values were continuously changed to get better fitting results, and some of the results are as follows.

When comparing the fitted curve with the field data (Fig. 11), it can

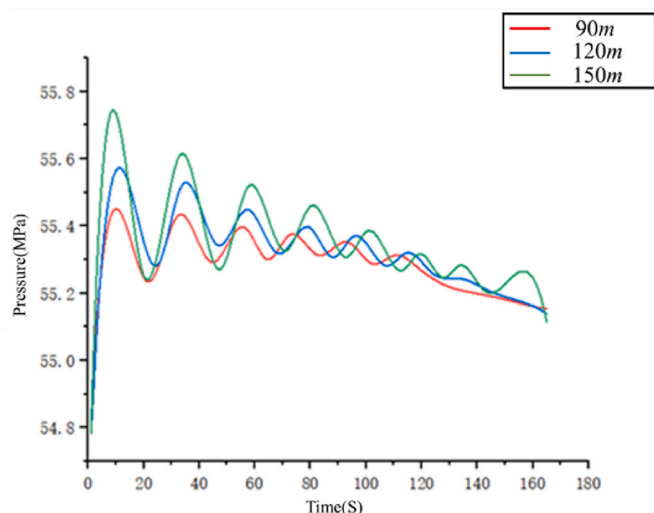


Fig. 7. Pressure curves for different segment lengths.

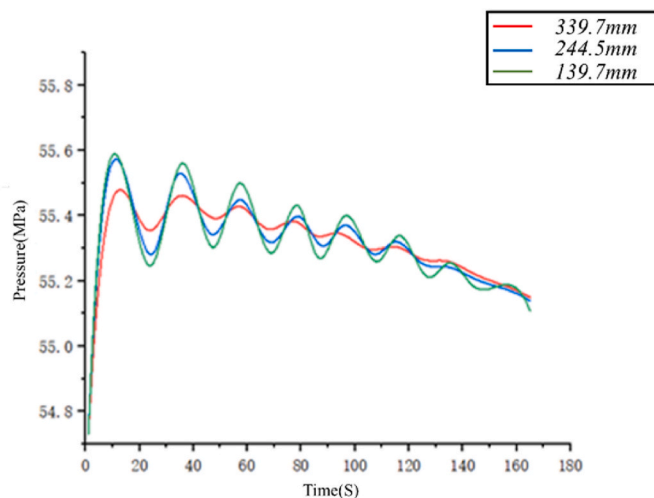


Fig. 8. Pressure curves for different casing sizes.

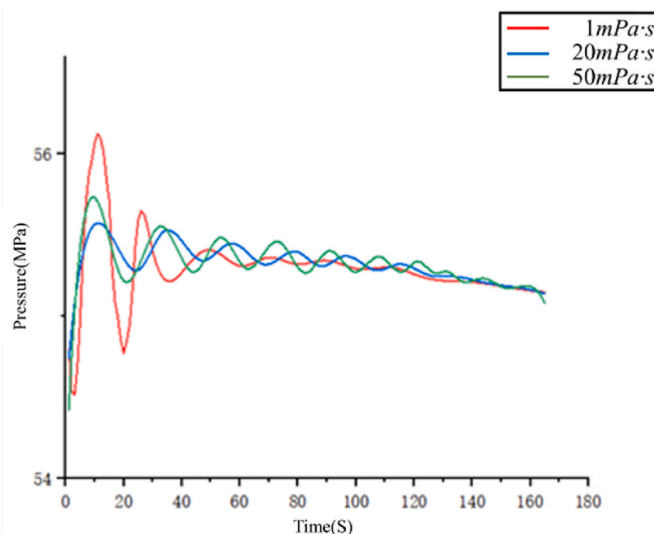


Fig. 9. Pressure curves for different fluid viscosities.

be observed that the decay degree, period, and amplitude of the curve align well. Nevertheless, it is important to acknowledge that during the initial phase of pump stoppage, the fitting results may exhibit larger errors due to factors such as the specific well shutdown method employed and potential interference from ground equipment during the data collection process.

The fracture size obtained from the simulation in this study is compared to the fracture size measured through microseismic monitoring, and selected results are presented in the following table (Table 2). The comparison reveals that the simulation results align more closely with reality and generally correspond to the field monitoring data. The average accuracy exceeds 80%, indicating the credibility of the stopping pressure analysis method based on the RCI model.

4. Conclusions

In this study, a novel method is developed to analyze pressure variations resulting from pump stoppage using the water hammer effect, which is not commonly utilized in field fracturing operations. Unlike traditional water hammer models, modifications are made to the original constant pressure boundary when dealing with the bottom hole boundary. The RCI circuit model is introduced to accurately describe the

Sand fracturing construction curve

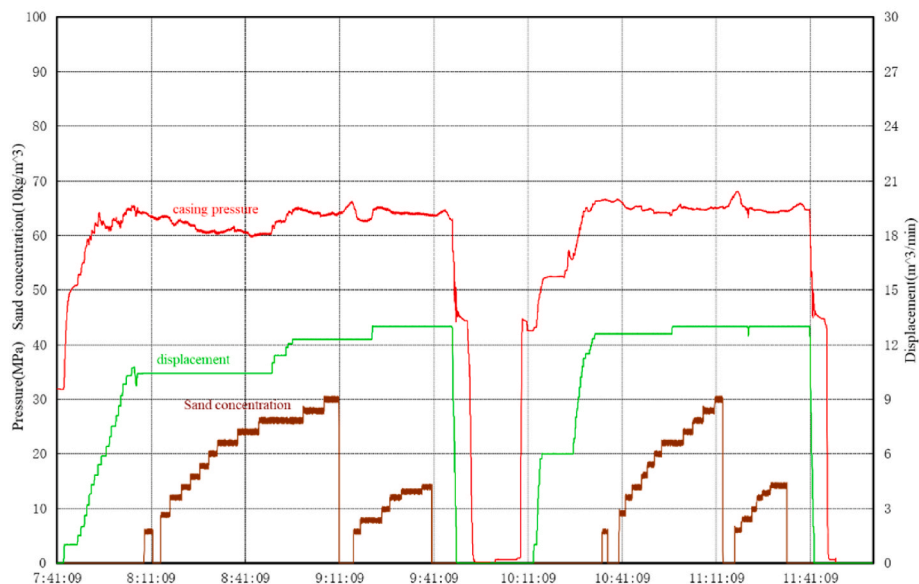


Fig. 10. Fracturing construction curve of a section of well 1.

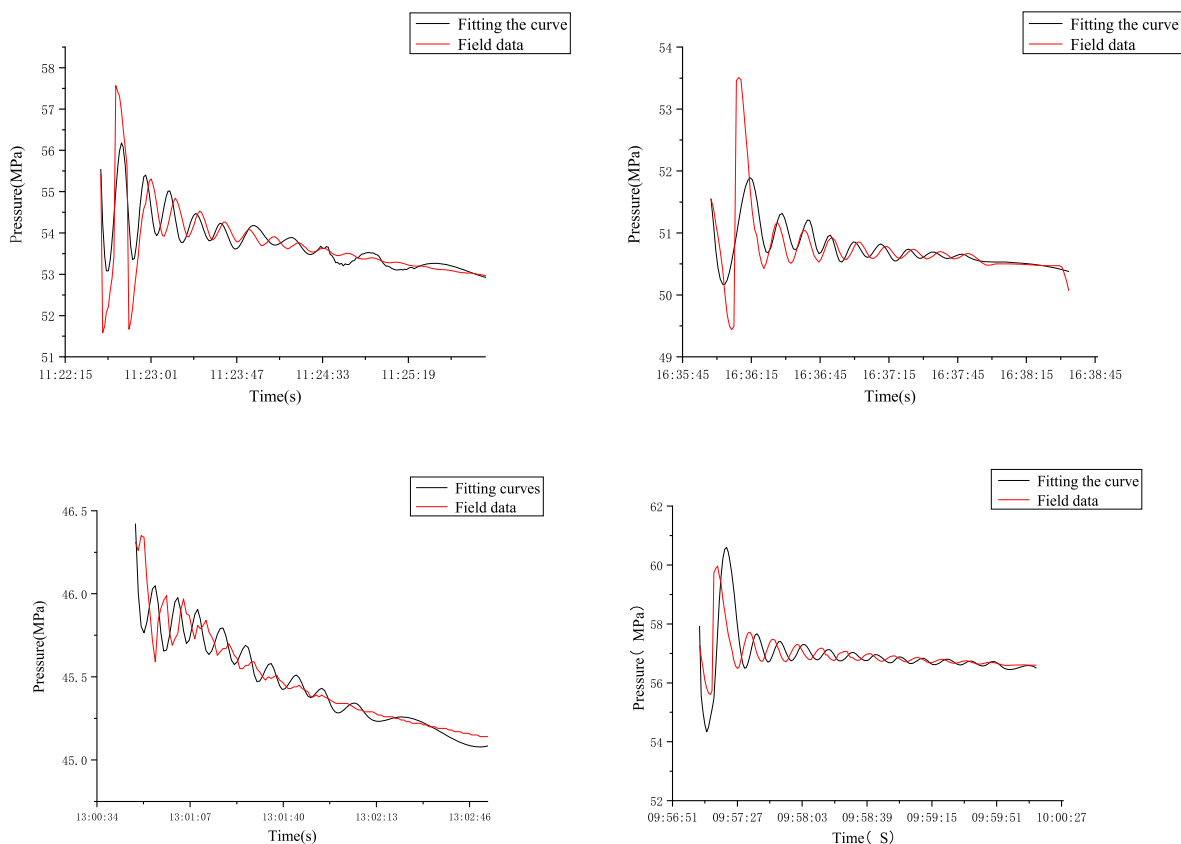


Fig. 11. Water hammer Pressure Curve.

bottom hole fracture, enabling efficient calculations while fully capturing the fluid and pressure changes within the fracture. By iteratively adjusting the parameters R , C , and I to match the field pressure data, the size parameters of the fracture can be accurately inverted based on circuit characteristics. Larger resistance R value indicates a smaller fracture volume. C value can be used to monitor any poorly performing

stages of the fracturing simulation process. I value mainly affects the parameter calculation of the fracture size (especially the fracture width).

The pressure analysis method is applied to analyze the water hammer effect following hydraulic fracturing stoppage in horizontal wells located in block W2. The accuracy of the fracture size parameters obtained from simulation, compared to microseismic monitoring data,

Table 2

Comparison of simulated fracture sizes and micro seismic results for single wells.

Well section	Parameters	Simulation	Microearthquake	Accuracy	Well section	Parameters	Simulation	Microearthquake	Accuracy
1	Fracture half-length	151.8	175	96%	10	Fracture half-length	153.2	155	93%
	Fracture height	37.69	40	94%		Fracture height	36.51	40	99%
2	Fracture half-length	164.5	185	90%	11	Fracture half-length	159.5	155	94%
	Fracture height	33.41	40	83%		Fracture height	39.06	50	73%
3	Fracture half-length	168.3	185	98%	12	Fracture half-length	152.8	165	98%
	Fracture height	44.89	40	95%		Fracture height	36.58	40	87%
4	Fracture half-length	167.8	160	97%	13	Fracture half-length	171.0	150	92%
	Fracture height	36.76	50	79%		Fracture height	36.08	40	83%
5	Fracture half-length	172.5	185	93%	14	Fracture half-length	164.4	165	98%
	Fracture height	35.78	40	89%		Fracture height	34.84	40	93%
6	Fracture half-length	172.6	195	81%	15	Fracture half-length	168.2	175	90%
	Fracture height	33.60	40	100%		Fracture height	38.33	40	86%
7	Fracture half-length	154.0	160	95%	16	Fracture half-length	153.6	150	95%
	Fracture height	42.4	50	70%		Fracture height	34.96	40	92%
8	Fracture half-length	150.6	190	83%	17	Fracture half-length	1960.8	150	94%
	Fracture height	39.33	40	93%		Fracture height	30.06	40	83%
9	Fracture half-length	175.4	185	85%					
	Fracture height	36.74	40	91%					

exceeds 80%. This validation demonstrates the accuracy of the pressure analysis method based on the RCI model for pump stoppage.

The model is further utilized for sensitivity analysis of the water impact effect, examining the influence of both RCI parameters and construction parameters. The findings provide valuable insights into the factors influencing the water impact effect and serve as a guide for achieving more accurate simulation results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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